Physical Chemistry of Materials

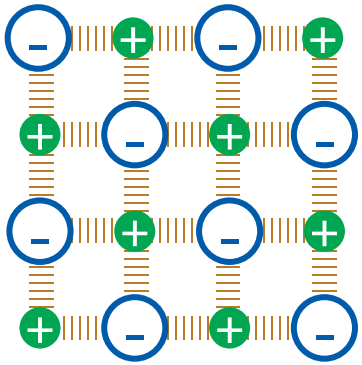
Lecture 11

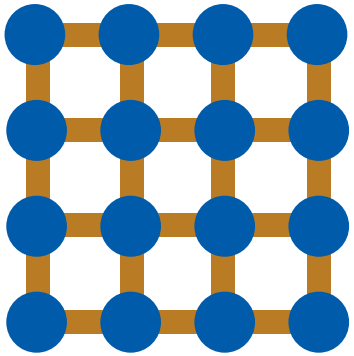
Crystal Vibrations: Classical Picture

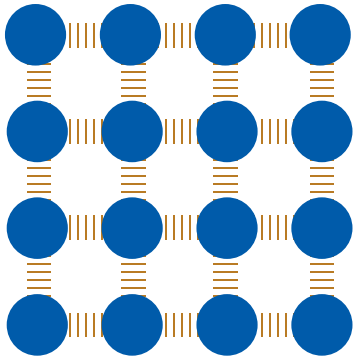
Last Class

Bonding in Solids

- Ionic solids: ionic bonds
- Covalent solids: covalent bonds
- Molecular solids: intermolecular interactions







Metals need more discussion

Last Class

Lattice Energy of Ionic Solids

- Lattice energy:

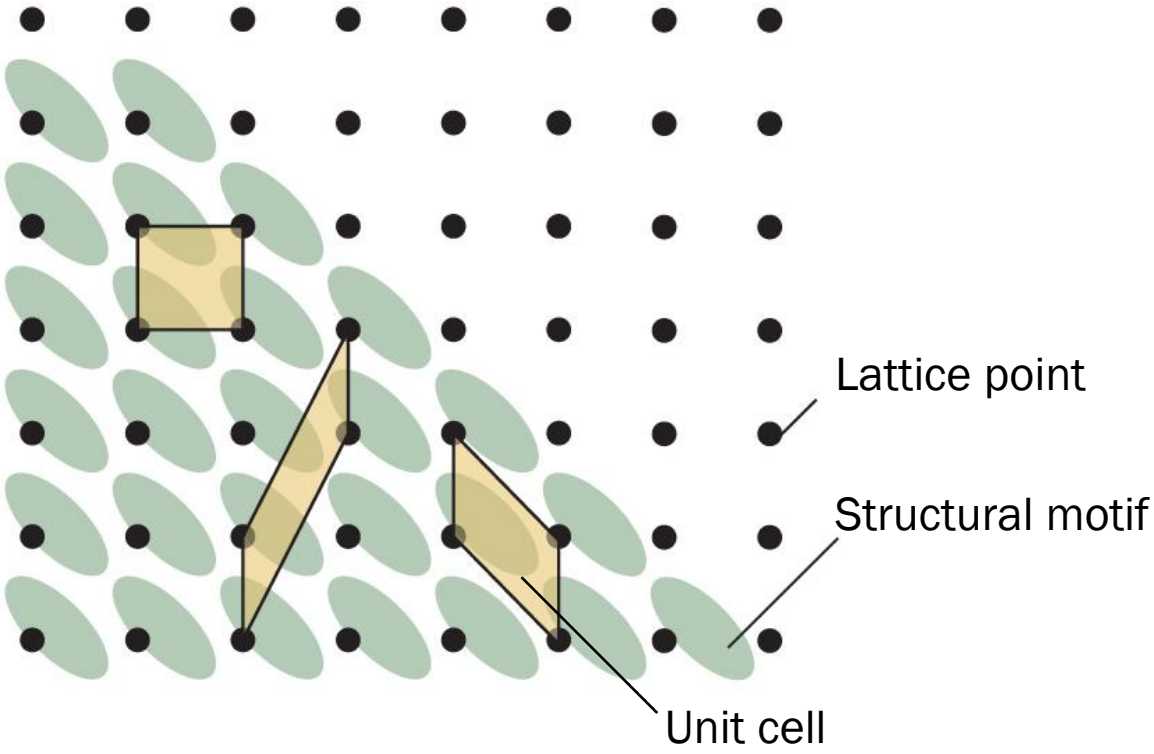
ΔU for MX(s) ⟶ M^+(g) + X^-(g)

- Two main contributions
  - Electrostatic interactions (attraction + repulsion)
  - Pauli repulsion
- Born-Meyer equation

$$E_{\mathrm{p,tot}}(d_{\mathrm{eq}}) = -A\frac{|z_{\mathrm{A}}z_{\mathrm{B}}|e^2}{4\pi\varepsilon_0 d_{\mathrm{eq}}}\left(1-\frac{d^*}{d_{\mathrm{eq}}}\right)$$

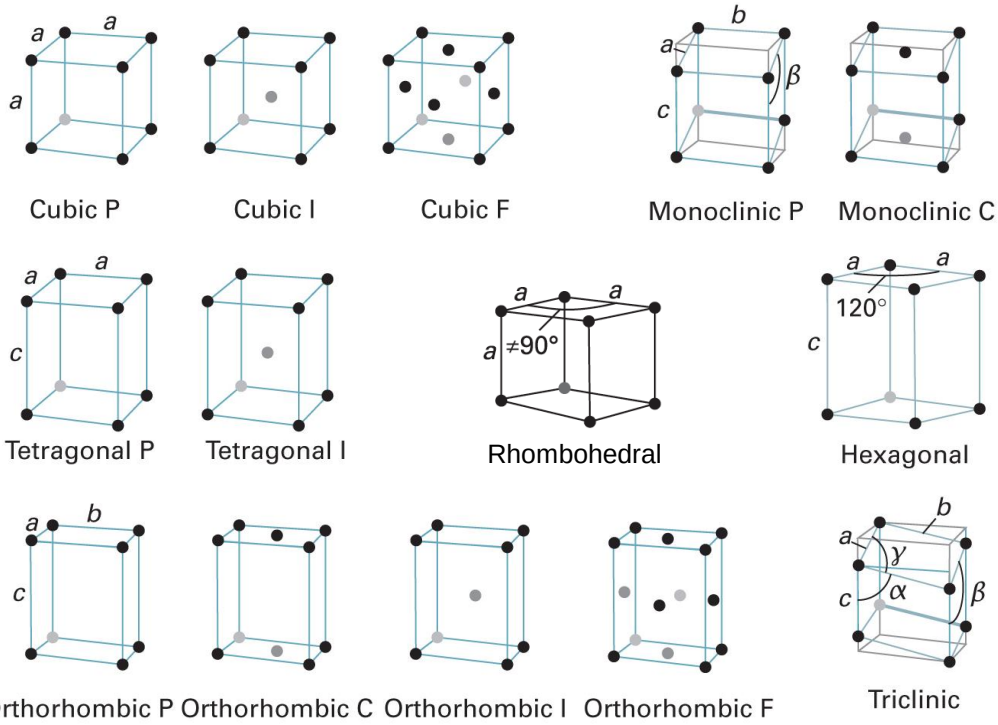
Unit Cells

whose purely translational displacements can span the entire space



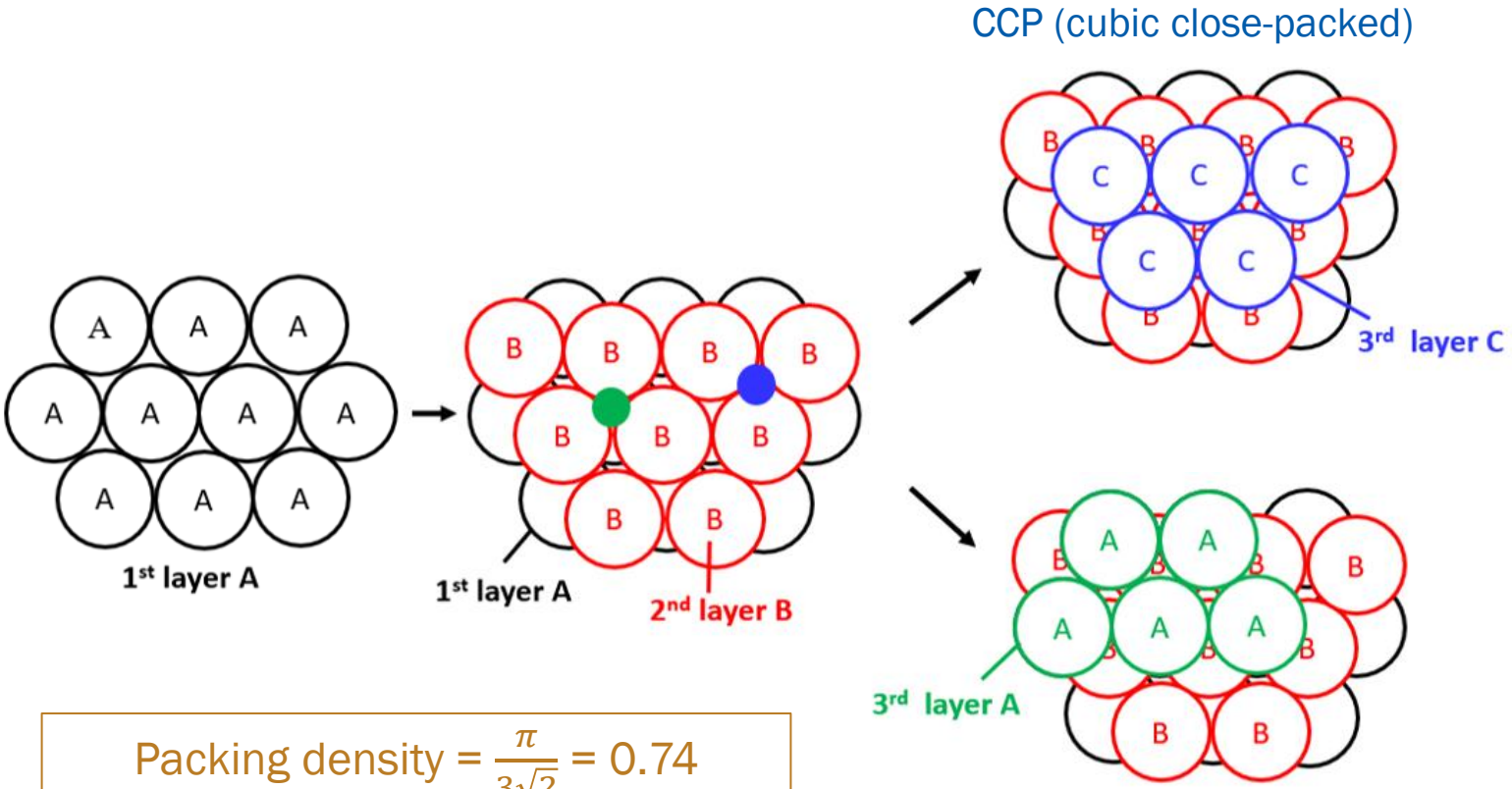
(Image: Atkins *et al.*, ***Atkins' Physical Chemistry***, 12th ed., Figures 15A.1 and 15A.2)

14 Bravais Lattices



(Image: Atkins *et al.*, ***Atkins' Physical Chemistry***, 12th ed., Figure 15A.8)

Close Packing

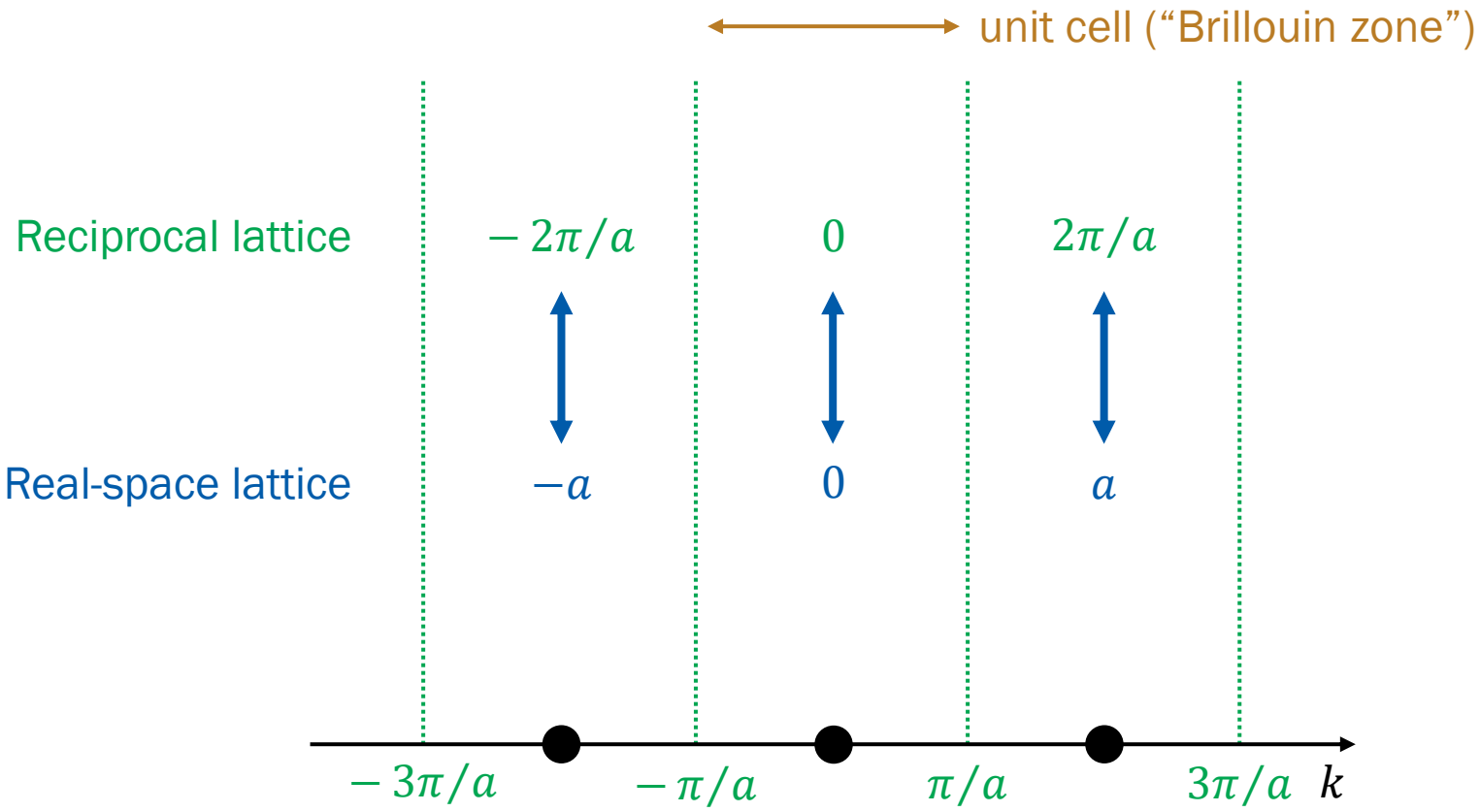


Packing density = $\frac{\pi}{3\sqrt{2}}$ = 0.74

(Kepler conjecture: proven in 1998)

(Image: https://commons.wikimedia.org/wiki/File:Cubic_Closest_Packing_(CCP)_and_Hexagonal_Closet_Packing_(HCP).png)

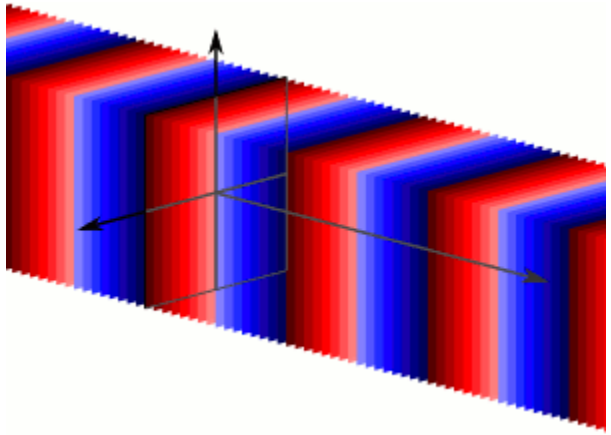
Reciprocal Lattice



3D Generalization

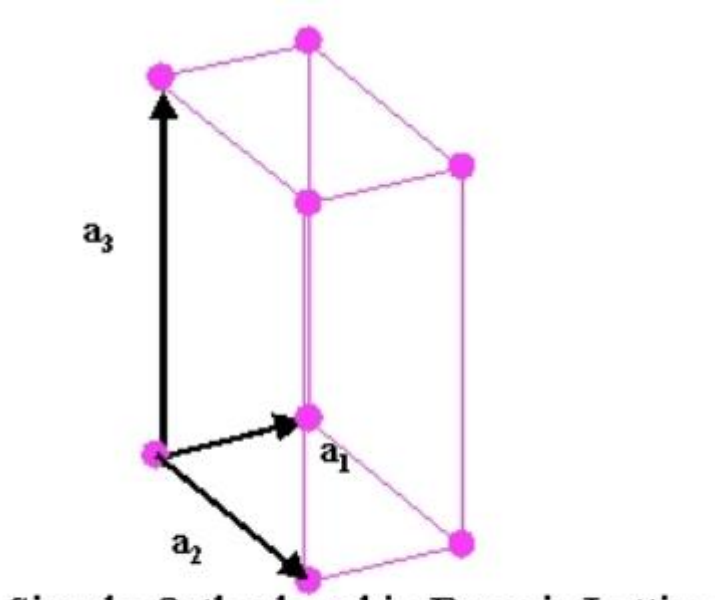


$$A(x,t) = A_0 \exp(ikx - i\omega t)$$

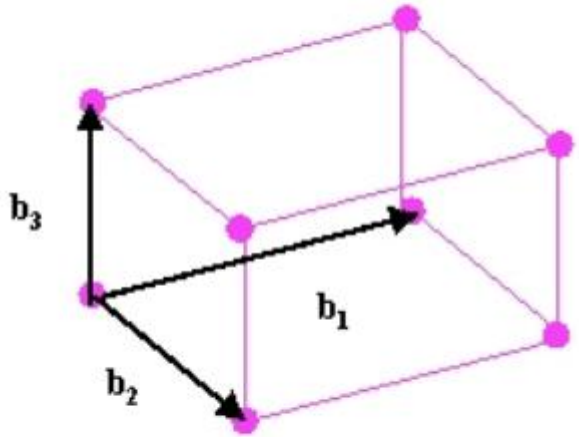


(Images: https://upload.wikimedia.org/wikipedia/commons/8/89/Travelling_wave_animated_plot.gif
https://commons.wikimedia.org/wiki/File:Plane_Wave_3D_Animation_300x216_255Colors.gif)

Reciprocal Lattice: Simple Case



Simple Orthorhombic Bravais Lattice
with $a_3 > a_2 > a_1$



Reciprocal Lattice
Note: $b_1 > b_2 > b_3$

(Image: https://www.physics.smu.edu/scalise/P6338sp16/albert/chapter2.pdf)

Reciprocal Lattice: General Case

For the real-space primitive lattice vectors $\vec{a}_1, \vec{a}_2, \vec{a}_3$, the primitive lattice vectors of the reciprocal lattice must satisfy

$$\vec{a}_i \cdot \vec{b}_j = 2\pi\delta_{ij}$$

which leads to

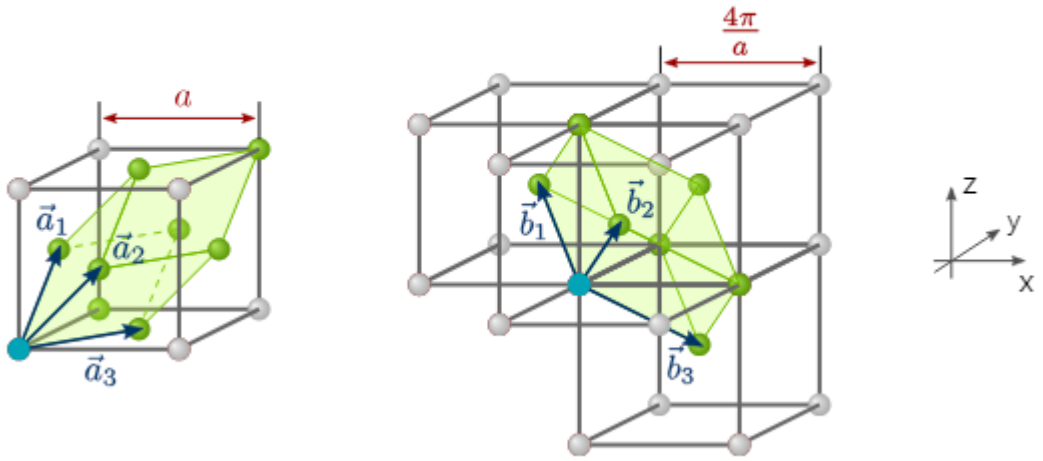
$$\vec{b}_1 = 2\pi \frac{\vec{a}_2 \times \vec{a}_3}{\vec{a}_1 \cdot (\vec{a}_2 \times \vec{a}_3)}; \quad \vec{b}_2 = 2\pi \frac{\vec{a}_3 \times \vec{a}_1}{\vec{a}_1 \cdot (\vec{a}_2 \times \vec{a}_3)};$$

$$\vec{b}_3 = 2\pi \frac{\vec{a}_1 \times \vec{a}_2}{\vec{a}_1 \cdot (\vec{a}_2 \times \vec{a}_3)}$$

Reciprocal Lattice: Example

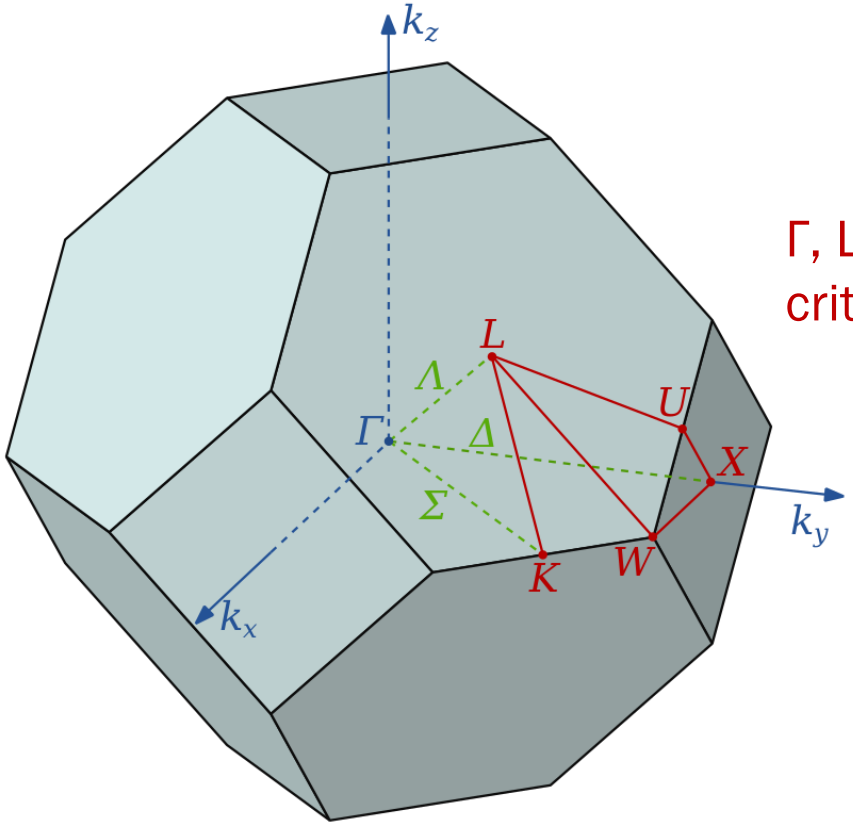
direct lattice:
fcc with edge length a

reciprocal lattice:
bcc with edge length 4π/a



(Image: https://www.physics-in-a-nutshell.com/article/15/the-reciprocal-lattice)

Brillouin Zone (FCC)

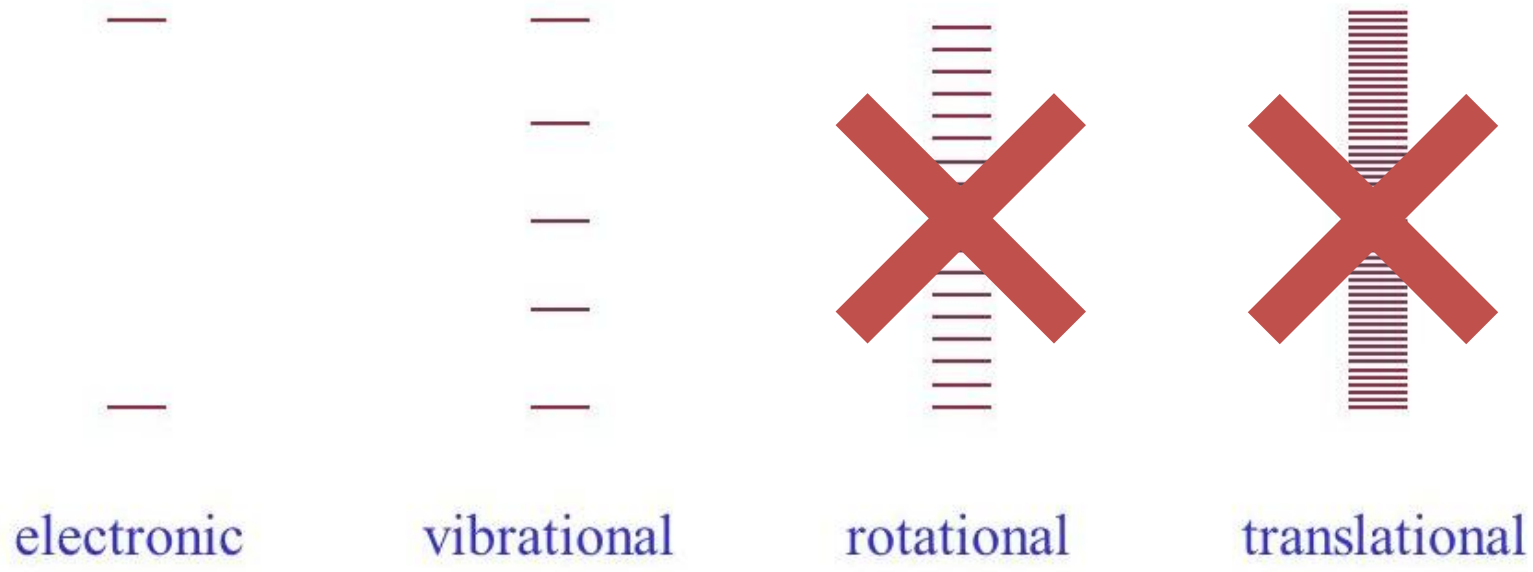


Γ, L, U, X, W, K:
critical points

(Image: https://commons.wikimedia.org/wiki/File:Brillouin_Zone_(1st,_FCC).svg)

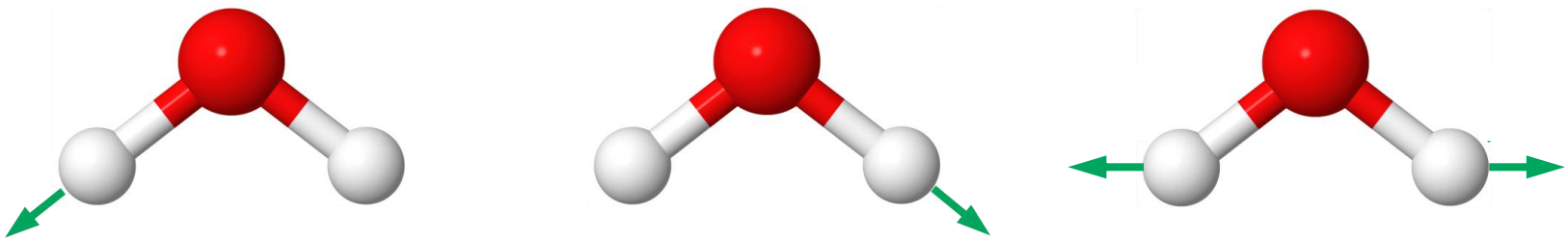
Crystal Energies

Molecular case:



(Image: https://slideplayer.com/slide/6282680/)

Molecular Vibration



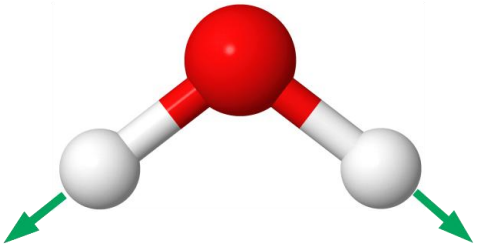
Center of mass is not fixed!

(coupled with translation)

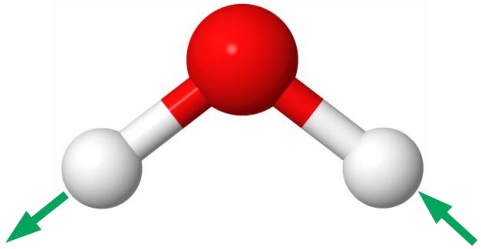
(Image: https://strawberryfields.ai/photonics/_images/water.png)

Normal Modes

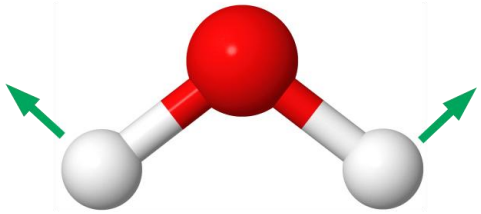
a pattern of motion in which all atoms move with the same frequency



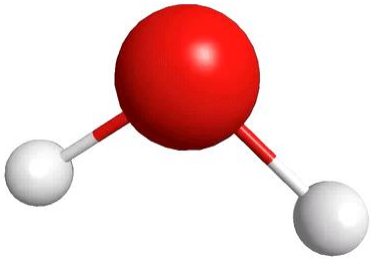
symmetric stretching



asymmetric stretching

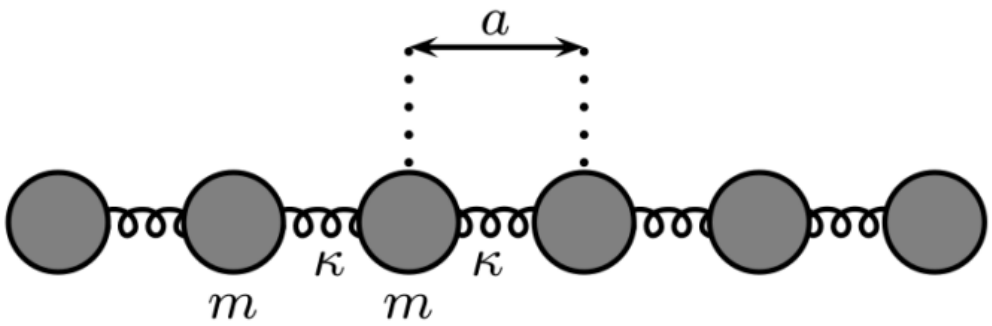


bending



(Images: https://strawberryfields.ai/photonics/_images/water.png
https://thepaperccny.com/2022/02/28/harmony-of-the-ocean/)

Normal Modes of 1D Crystal



- Deviation from equilibrium position $q_n = x_n - x_{n0}$
- Total potential energy

$$V_{\text{tot}} = \sum_{i=1}^{N} \frac{\kappa}{2} (x_{i+1} - x_i - a)^2 = \sum_{i=1}^{N} \frac{\kappa}{2} (q_{i+1} - q_i)^2$$

(Image: Simon, *The Oxford Solid State Basics* (2013), Fig. 9.1)

Newton's Law

$$V_{\text{tot}} = \sum_{i=1}^{N} \frac{\kappa}{2}(x_{i+1} - x_i - a)^2 = \sum_{i=1}^{N} \frac{\kappa}{2}(q_{i+1} - q_i)^2$$

- Force on particle n

$$F_n = -\frac{\partial V_{\text{tot}}}{\partial x_n} = \kappa(q_{n+1} - q_n) + \kappa(q_{n-1} - q_n)$$

$$F_n = ma_n = m\frac{\partial^2 q_n}{\partial t^2} = \kappa(q_{n+1} + q_{n-1} - 2q_n)$$

Solving the Differential Equation

$$m\frac{\partial^2 q_n}{\partial t^2} = \kappa(q_{n+1} + q_{n-1} - 2q_n)$$

- Ansatz ("educated guess"):

$$q_n = A\exp(i\omega t - ikx_{n0}) = A\exp(i\omega t - ikna)$$

Substitution leads to

$$-m\omega^2 A e^{i\omega t - ikna} = \kappa A\left(e^{i\omega t - ik(n+1)a} + e^{i\omega t - ik(n-1)a} - 2e^{i\omega t - ikna}\right)$$

$$-m\omega^2 = \kappa\left(e^{-ika} + e^{ika} - 2\right)$$

$$m\omega^2 = 4\kappa\sin^2(ka/2)$$

Normal Mode Frequency

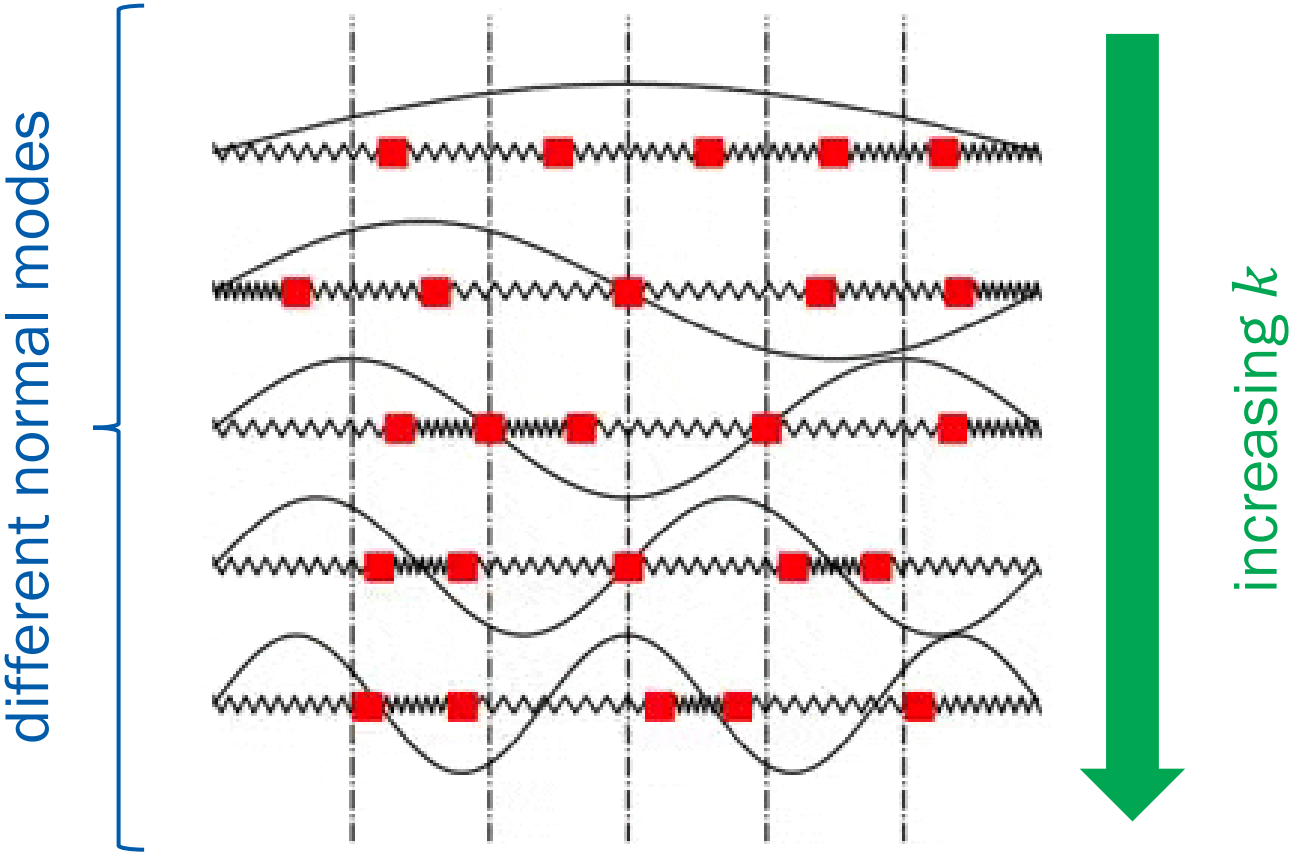
$$m\omega^2 = 4\kappa \sin^2(ka/2)$$

Thus,

$$\omega = 2\sqrt{\frac{\kappa}{m}}\left|\sin\left(\frac{ka}{2}\right)\right|$$

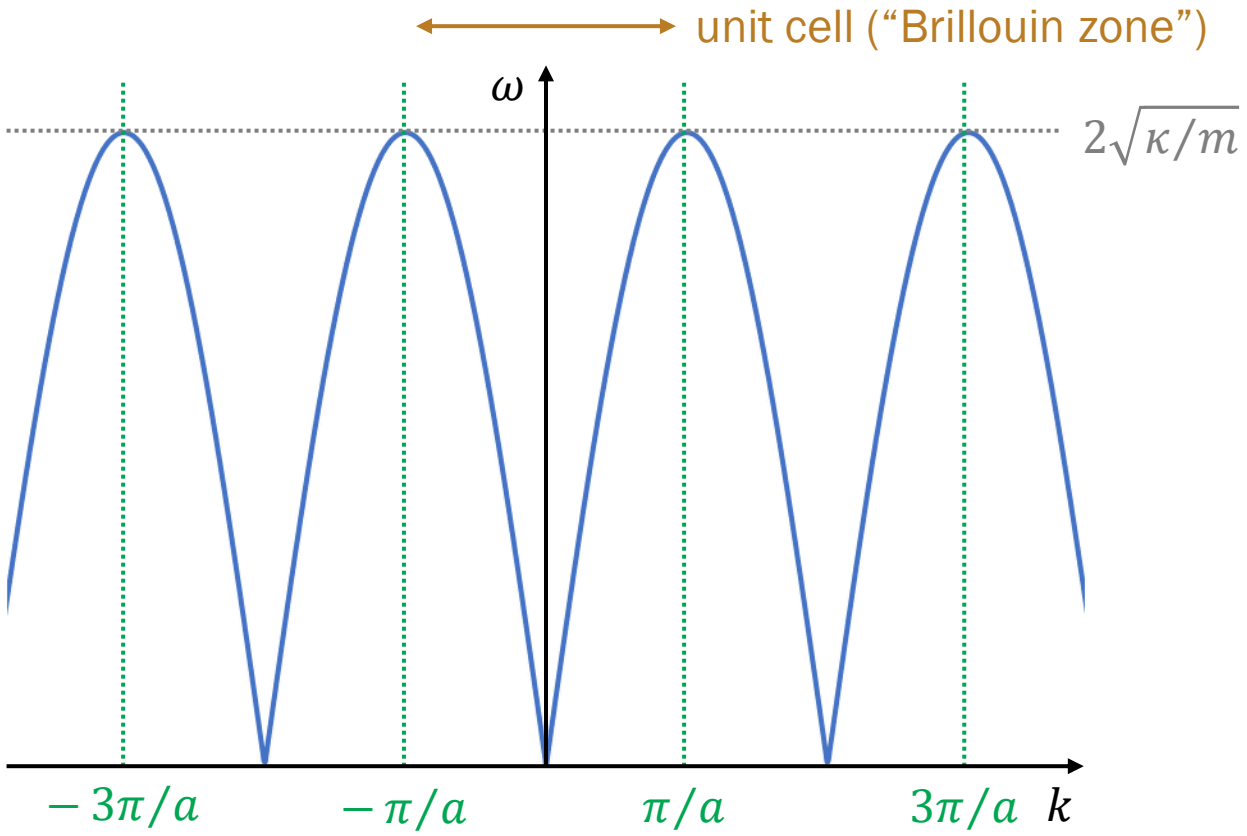
(dispersion relation)

Lattice Vibrations

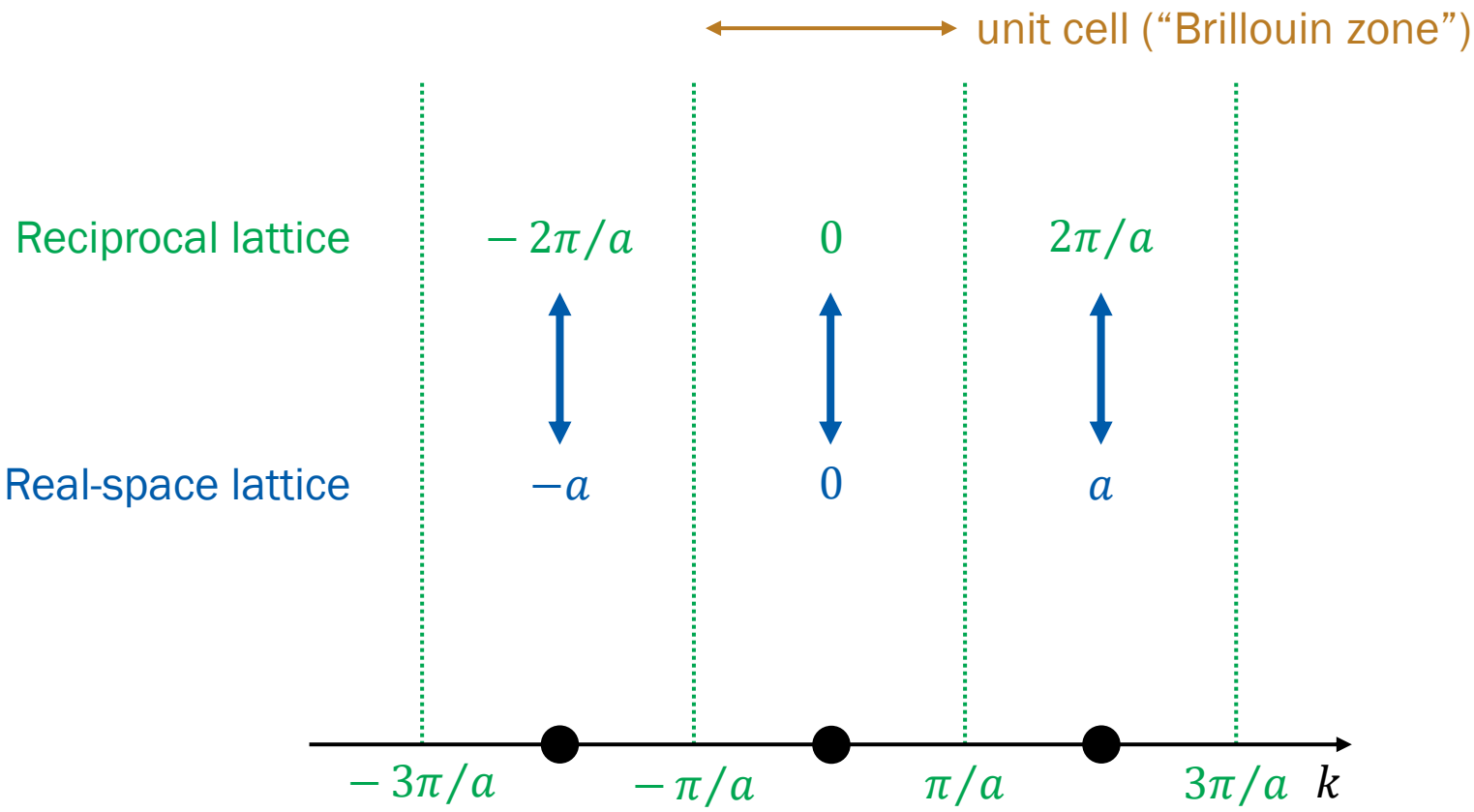


(Image: https://www.quora.com/What-is-an-intuitive-explanation-of-phonons)

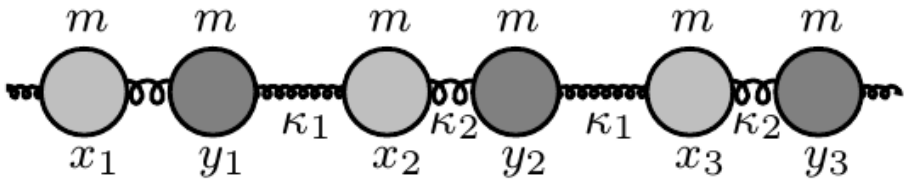
Dispersion Relation



Reciprocal Lattice



Case of Diatomic Solid



Define $q_n = x_n - x_{n0}$ and $q'_n = y_n - y_{n0}$;

$$V_{\text{tot}} = \sum_{i=1}^{N} \frac{\kappa_1}{2}(x_{i+1} - y_i - a)^2 + \sum_{i=1}^{N} \frac{\kappa_2}{2}(y_i - x_i - a)^2$$

$$V_{\text{tot}} = \sum_{i=1}^{N} \left[\frac{\kappa_1}{2}(q_{i+1} - q'_i)^2 + \frac{\kappa_2}{2}(q'_i - q_i)^2\right]$$

(Image: Simon, *The Oxford Solid State Basics* (2013), Fig. 10.5)

Forces on Two Types of Particles

$$V_{\text{tot}} = \sum_{i=1}^{N} \left[\frac{\kappa_1}{2} (q_{i+1} - q_i')^2 + \frac{\kappa_2}{2} (q_i' - q_i)^2 \right]$$

$$F_n = -\frac{\partial V_{\text{tot}}}{\partial x_n} = -\frac{\partial V_{\text{tot}}}{\partial q_n} = -\kappa_1 (q_n - q_{n-1}') + \kappa_2 (q_n' - q_n)$$

$$F_n' = -\frac{\partial V_{\text{tot}}}{\partial y_n} = -\frac{\partial V_{\text{tot}}}{\partial q_n'} = \kappa_1 (q_{n+1} - q_n') - \kappa_2 (q_n' - q_n)$$

Newton's Law

$$F_n = -\kappa_1(q_n - q'_{n-1}) + \kappa_2(q'_n - q_n)$$

$$F'_n = \kappa_1(q_{n+1} - q'_n) - \kappa_2(q'_n - q_n)$$

As $F = ma = m(\partial^2 x/\partial t^2)$,

$$m\frac{\partial^2 q_n}{\partial t^2} = -\kappa_1(q_n - q'_{n-1}) + \kappa_2(q'_n - q_n)$$

$$m\frac{\partial^2 q'_n}{\partial t^2} = \kappa_1(q_{n+1} - q'_n) - \kappa_2(q'_n - q_n)$$

Solving the Differential Equations

$$m\frac{\partial^2 q_n}{\partial t^2} = -\kappa_1(q_n - q'_{n-1}) + \kappa_2(q'_n - q_n)$$

$$m\frac{\partial^2 q'_n}{\partial t^2} = \kappa_1(q_{n+1} - q'_n) - \kappa_2(q'_n - q_n)$$

- Ansätze: $q_n = A_x e^{i\omega t - ikna}$ and $q'_n = A_y e^{i\omega t - ikna}$

$$-m\omega^2 A_x = \kappa_2 A_y + \kappa_1 A_y e^{ika} - (\kappa_1 + \kappa_2)A_x$$

$$-m\omega^2 A_y = \kappa_1 A_x e^{-ika} + \kappa_2 A_x - (\kappa_1 + \kappa_2)A_y$$

Eigenvalue Problem

$$-m\omega^2 A_x = \kappa_2 A_y + \kappa_1 A_y e^{ika} - (\kappa_1 + \kappa_2) A_x$$

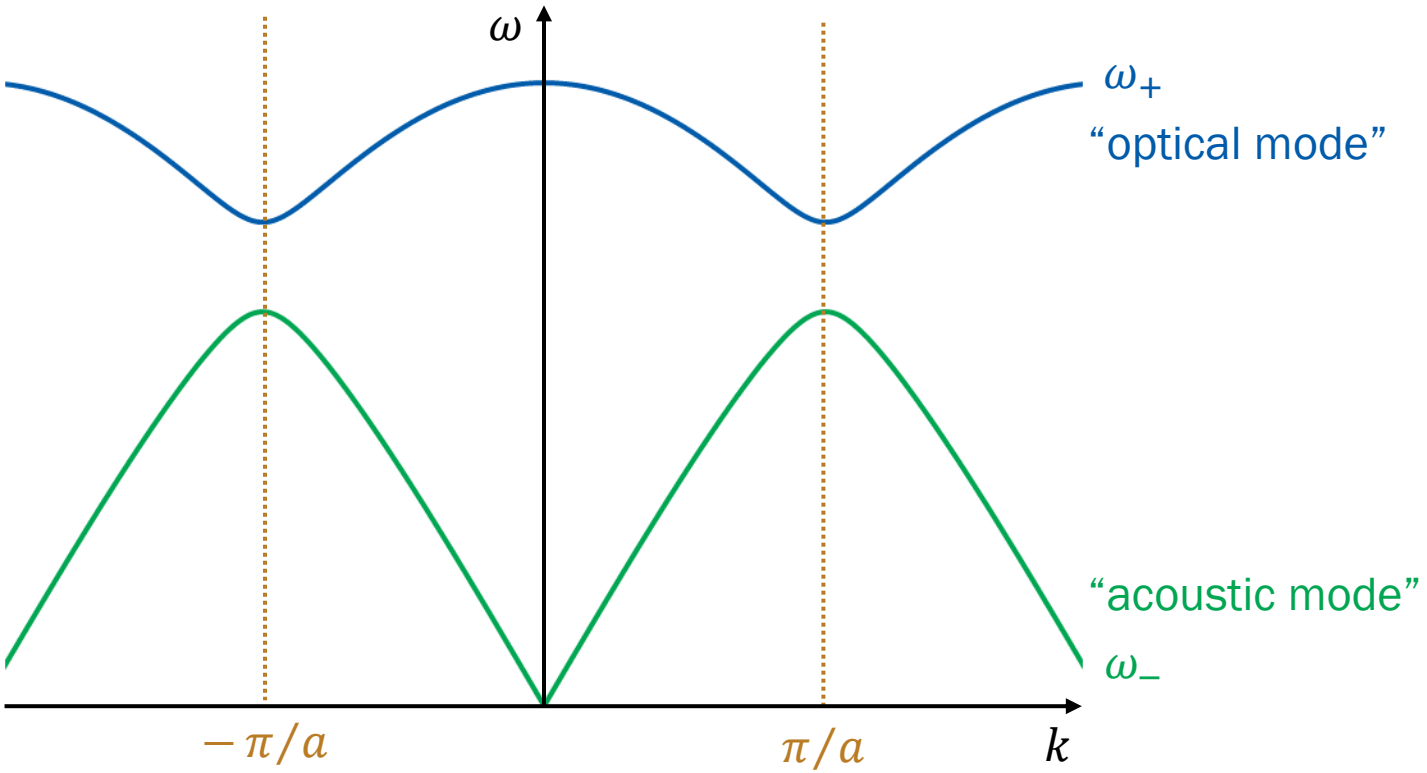
$$-m\omega^2 A_y = \kappa_1 A_x e^{-ika} + \kappa_2 A_x - (\kappa_1 + \kappa_2) A_y$$

$$m\omega^2 \begin{pmatrix} A_x \\ A_y \end{pmatrix} = \begin{pmatrix} \kappa_1 + \kappa_2 & -\kappa_2 - \kappa_1 e^{ika} \\ -\kappa_2 - \kappa_1 e^{-ika} & \kappa_1 + \kappa_2 \end{pmatrix} \begin{pmatrix} A_x \\ A_y \end{pmatrix}$$

Solving the eigenvalue problem, we obtain

$$\omega_{\pm} = \sqrt{\frac{\kappa_1 + \kappa_2}{m} \pm \frac{1}{m}\sqrt{\kappa_1^2 + \kappa_2^2 + 2\kappa_1\kappa_2 \cos(ka)}}$$

Dispersion Relation



Nature of the Two Modes

$$m\omega^2 \begin{pmatrix} A_x \\ A_y \end{pmatrix} = \begin{pmatrix} \kappa_1 + \kappa_2 & -\kappa_2 - \kappa_1 e^{ika} \\ -\kappa_2 - \kappa_1 e^{-ika} & \kappa_1 + \kappa_2 \end{pmatrix} \begin{pmatrix} A_x \\ A_y \end{pmatrix}$$

Take $k \to 0$;

$$m\omega^2 \begin{pmatrix} A_x \\ A_y \end{pmatrix} = \begin{pmatrix} \kappa_1 + \kappa_2 & -\kappa_2 - \kappa_1 \\ -\kappa_2 - \kappa_1 & \kappa_1 + \kappa_2 \end{pmatrix} \begin{pmatrix} A_x \\ A_y \end{pmatrix}$$

$$\omega^2 \begin{pmatrix} A_x \\ A_y \end{pmatrix} = \frac{\kappa_1 + \kappa_2}{m} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} A_x \\ A_y \end{pmatrix}$$

Nature of the Two Modes

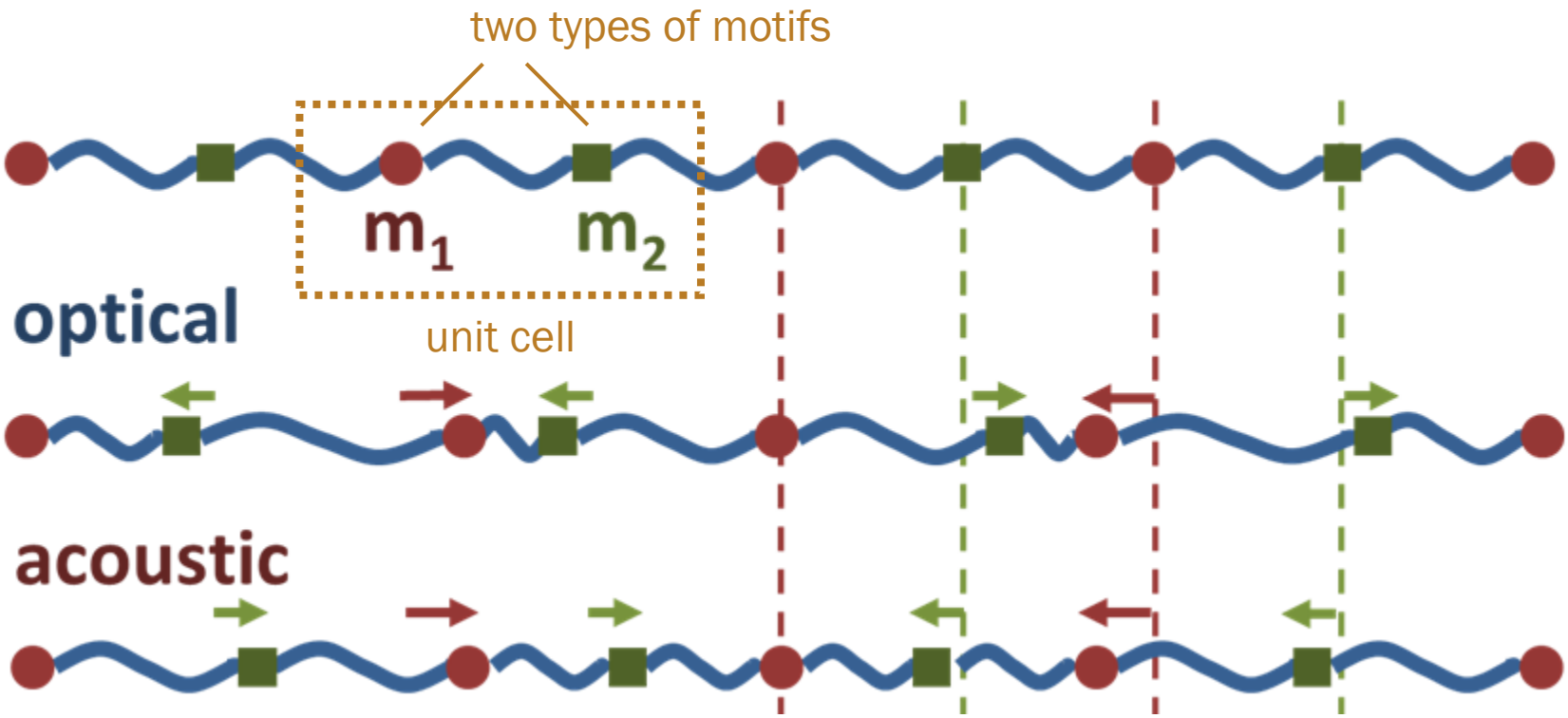
$$\omega^2 \begin{pmatrix} A_x \\ A_y \end{pmatrix} = \frac{\kappa_1 + \kappa_2}{m} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} A_x \\ A_y \end{pmatrix}$$

Solving the eigenvalue problem, the eigenvectors corresponding to ω_+ and ω_- are

$$\begin{pmatrix} A_x \\ A_y \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix} \text{ and } \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

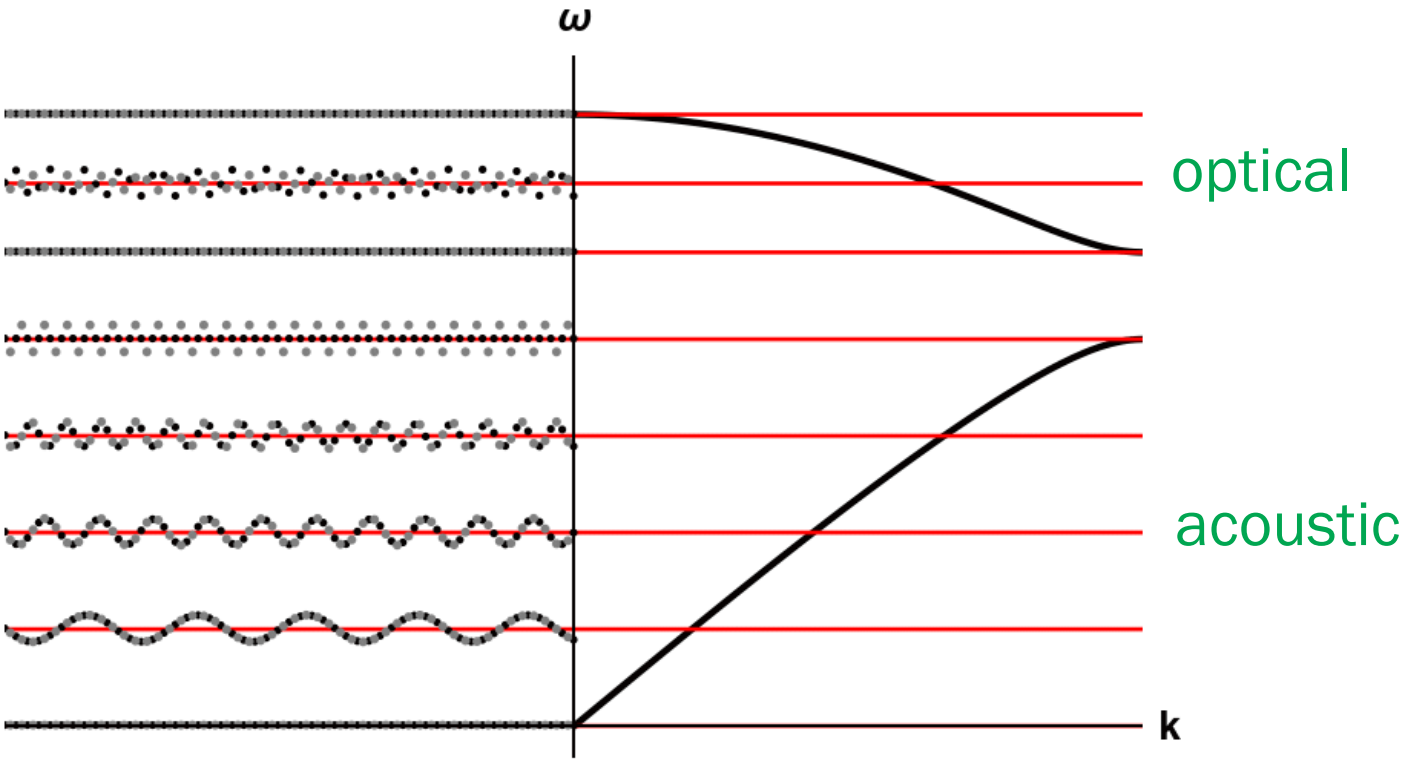
(Recall that $q_n = A_x e^{i\omega t - ikna}$ and $q'_n = A_y e^{i\omega t - ikna}$)

Acoustic and Optical Phonons



(Image: https://commons.wikimedia.org/wiki/File:Optical_%26_acoustic_vibrations.png)

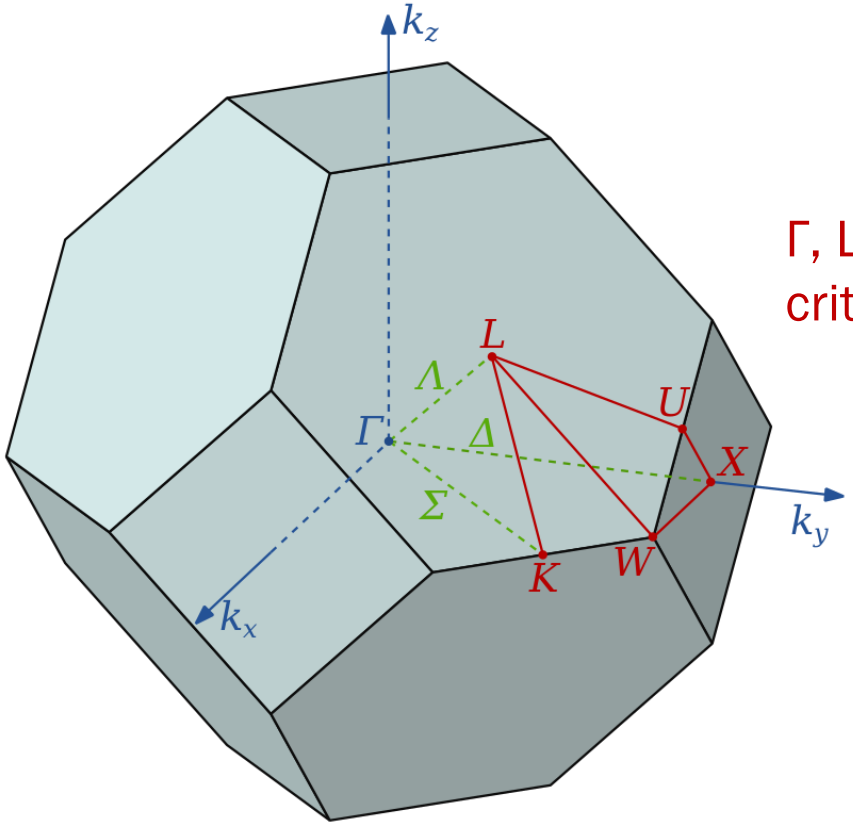
Dispersion Relation



(Image: https://upload.wikimedia.org/wikipedia/commons/d/d7/Diatomic_chain.gif)

Now, Expand to 3D Systems!

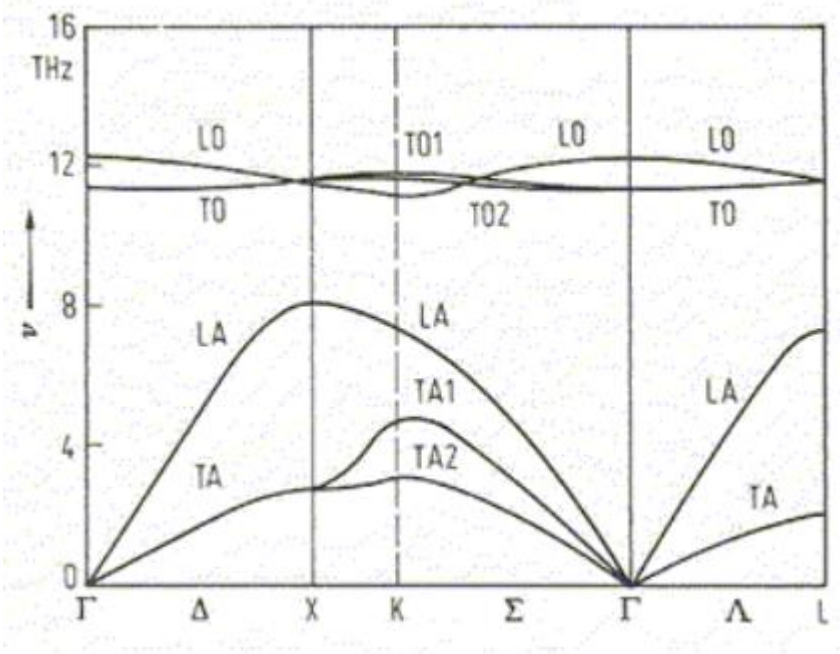
Brillouin Zone (FCC)

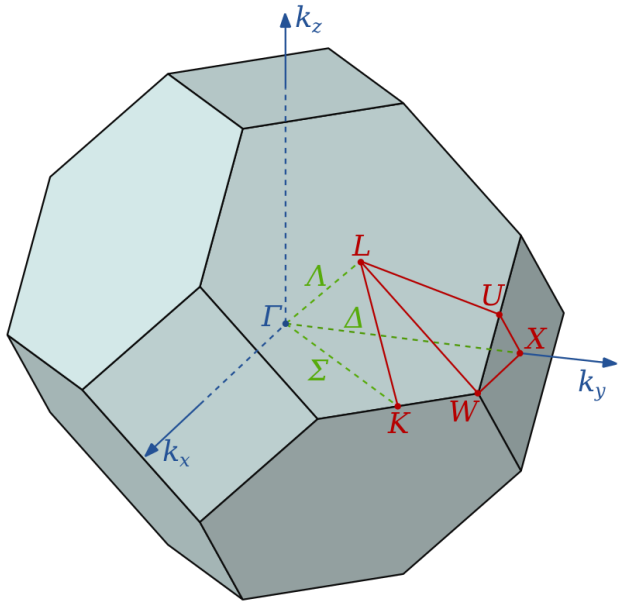


Γ, L, U, X, W, K:
critical points

(Image: https://commons.wikimedia.org/wiki/File:Brillouin_Zone_(1st,_FCC).svg)

Dispersion Relation (FCC)





(Images: http://lampz.tugraz.at/~hadley/ss1/problems/problems_phonons.php
https://commons.wikimedia.org/wiki/File:Brillouin_Zone_(1st,_FCC).svg)